

Age-related macular degeneration: a leading cause of blindness.

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Age-related macular degeneration (AMD) is the most common cause of blindness in Australia, rising in prevalence from zero in people aged under 55 years to 18.5% in those aged over 85 years. More women suffer from the disease and it is estimated that 72,220 Australians may have signs of late AMD in one or both eyes. Degenerative processes in the retina include the appearance of abnormal drüsen (the key lesion of AMD); atrophy, retinal-pigment abnormalities and gradual loss of vision (atrophic AMD); and haemorrhage, exudation, scarring and rapid loss of vision (exudative AMD). Although environmental (e.g., smoking) and hereditary factors may possibly play a role in the development of AMD, the principal risk factor is ageing. Treatment remains supportive for most patients, but a few will benefit from macular laser photocoagulation.

Age-related macular degeneration: A two-level model hypothesis.

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Age-related diseases, including age-related macular degeneration (AMD), are of growing importance in a world where population ageing has become a dominant global trend. Although a wide variety of risk factors for AMD have been identified, age itself remains by far the most important risk factor, making it an urgent priority to understand the connections between underlying ageing mechanisms and pathophysiology of AMD. Ageing is both multicausal and variable, so that differences between individuals in biological ageing processes are the focus of a growing number of pathophysiological studies seeking to explain how ageing contributes to chronic, age-related conditions. The aim of this review is to integrate the available knowledge on the pathophysiology of AMD within the framework of the biology of ageing. One highly significant feature of biological ageing is systemic inflammation, which arises as a second-level response to a first level of molecular damage involving oxidative stress, mutations etc. Combining these insights, the various co-existing pathophysiological explanations in AMD arrange themselves according to a two-level hypothesis. Accordingly, we describe how AMD can be considered the consequence of age-related random accumulation of molecular damage at the ocular level and the subsequent systemic inflammatory host response thereof. We summarize evidence and provide original data to enlighten where evidence is lacking. Finally, we discuss how this two-level hypothesis provides a foundation for thoughts and future studies in prevention, prognosis, and intervention.

Aging is not a disease: distinguishing age-related macular degeneration from aging.

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Age-related macular degeneration (AMD) is a disease of the outer retina, characterized most significantly by atrophy of photoreceptors and retinal pigment epithelium accompanied with or without choroidal neovascularization. Development of AMD has been recognized as contingent on environmental and genetic risk factors, the strongest being advanced age. In this review, we highlight pathogenic changes that destabilize ocular homeostasis and promote AMD development. With normal aging, photoreceptors are steadily lost, Bruch's membrane thickens, the choroid thins, and hard drusen may form in the periphery. In AMD, many of these changes are exacerbated in addition to the development of

disease-specific factors such as soft macular drusen. Para-inflammation, which can be thought of as an intermediate between basal and robust levels of inflammation, develops within the retina in an attempt to maintain ocular homeostasis, reflected by increased expression of the anti-inflammatory cytokine IL-10 coupled with shifts in macrophage plasticity from the pro-inflammatory M1 to the anti-inflammatory M2 polarization. In AMD, imbalances in the M1 and M2 populations together with activation of retinal microglia are observed and potentially contribute to tissue degeneration. Nonetheless, the retina persists in a state of chronic inflammation and increased expression of certain cytokines and inflammasomes is observed. Since not everyone develops AMD, the vital question to ask is how the body establishes a balance between normal age-related changes and the pathological phenotypes in AMD.

Retinal pigment epithelium and age-related macular degeneration: A review of major disease mechanisms.

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Age-related macular degeneration (AMD) is a progressive degenerative disease that is the leading cause of vision loss in the elderly population. Degeneration/dysregulation of the retinal pigment epithelium (RPE), a supportive monolayer of cells underlying the photoreceptors, is commonly seen in patients with AMD. While treatment exists for the neovascular/wet form of AMD, there is currently no cure for the non-exudative/dry form of AMD, making it imperative to understand the pathogenesis of this disease. Although our understanding of the aetiology of AMD has increased over the years, the underlying disease mechanism has not yet been identified, mainly due to the multifactorial nature of this disease. Herein, we review some of the commonly proposed degeneration pathways of RPE cells and their role in the pathogenesis of AMD; including activation of the complement cascade, oxidative stress-induced cell death mechanisms, dysfunctional mitochondria and the role of crystallins in AMD disease progression.

[New findings on pathogenetic mechanisms in the development of age-related macular degeneration].

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Age-related macular degeneration (AMD) is a complex multifactorial disease that occurs due to disfunction and degeneration of retinal pigment epithelium (RPE) and choriocapillaris, as well as death of photoreceptors. The exact pathogenetic mechanism remains uncertain. The aging process is the main and the clearest risk factor of AMD. In the development of this condition, a special role belongs to the secretory phenotype of aging spreading from one cell to another and mediated by the secretion and release of growth factors, cytokines, chemokines, proteases, and other molecules. Another major contributor is oxidative stress caused by violations in the recirculation of vitamin A in the vision cycle and accompanied by accumulation of lipofuscin, which mediates the formation of iron-based oxidants that are toxic for mitochondria. Furthermore, prolonged oxidative stress and constant light exposure induce the development of inflammation in the retina. Accumulation of metabolic products and cellular defects with age can induce an inflammatory reaction that amplifies the damage. The inflammatory processes including innate immune response, activation of microglia and parainflammation that occur locally in the vascular membrane, pigment epithelium and neuroretina are very significant contributors to the age-related changes, their progression, and the development of advanced stages of AMD. Various growth

factors play a special role in the development of choroidal neovascularization (CNV). Vascular endothelial growth factor A (VEGF-A) has traditionally been considered the main factor of neoangiogenesis and, consequently, the main therapeutic target, but in recent years various studies have determined the role of other factors - VEGF-B, C, D, PGF, Gal-1, angiopoietins. This article describes the main underlying mechanisms in the development of choroidal neovascularization including retinal aging, impaired metabolic activity, mitochondrial dysfunction, inflammatory reactions and genetic variations, as well as the role of various growth factors.

Does senescence play a role in age-related macular degeneration?

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Advanced age is the most established risk factor for developing age-related macular degeneration (AMD), one of the leading causes of visual impairment in the elderly, in Western and developed countries. Similarly, after middle age, there is an exponential increase in pathologic molecular and cellular events that can induce senescence, traditionally defined as an irreversible loss of the cells' ability to divide and most recently reported to also occur in select post-mitotic and terminally differentiated cells, such as neurons. Together these facts raise the question as to whether or not cellular senescence, may play a role in the development of AMD. A number of studies have reported the effect of ocular-relevant inducers of senescence using primarily in vitro models of poorly polarized, actively dividing retinal pigment epithelial (RPE) cell lines. However, in interpreting the data, the fidelity of these culture models to the RPE in vivo, must be considered. Fewer studies have explored the presence and/or impact of senescent cells in in vivo models that present with phenotypic features of AMD, leaving this an open field for further investigation. The goal of this review is to discuss current thoughts on the potential role of senescence in AMD development and progression, with consideration of the model systems used and their relevance to human disease.

Gut microbiota and age-related macular degeneration: A growing partnership.

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Age-related macular degeneration (AMD) is a leading cause of severe, irreversible vision impairment in developed countries, and its prevalence is rising all over the world, increasing sharply with age. AMD represents an acquired degeneration of the retina that causes significant central visual impairment through a combination of noneovascular and neovascular derangement. The main risk factors for the development of advanced AMD are increasing age, genetic factors, and cigarette smoking; however, the exact pathophysiology of AMD is yet relatively poorly understood. In recent years, the gut microbiota has been intensively studied and linked to several pathologic processes, including ocular diseases. In this sense, the aim of this review is to gather published evidence about the relationship between gut microbiota and AMD.

The role of inflammation in the pathogenesis of age-related macular degeneration.

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Age-related macular degeneration (AMD), the leading cause of blindness in the elderly, is a complex disease to study because of the potential role of demographic, environmental, and other systemic risk factors, such as age, sex, race, light exposure, diet, smoking, and underlying cardiovascular disease which may contribute to the pathogenesis of this disease. Recently, single nucleotide polymorphisms, DNA sequence variations found within the complement Factor H gene, have been found to be strongly associated with the development of AMD in Caucasians. One single nucleotide polymorphism, Tyr402His, was associated with approximately 50% of AMD cases. We review recent developments in the molecular biology of AMD, including single nucleotide polymorphisms within the Factor H gene, which may predispose individuals to the susceptibility of AMD as well as single nucleotide polymorphisms that may confer a protective effect. Taken together these findings help to provide new insights into the central issues surrounding the pathogenesis of AMD.

P38 MAPK Signaling in the Retina: Effects of Aging and Age-Related Macular Degeneration.

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Age-related macular degeneration (AMD) is the leading cause of irreversible visual impairment worldwide. Age is the greatest risk factor for AMD but the underlying mechanism remains unascertained, resulting in a lack of effective therapies. Growing evidence shows that dysregulation of the p38 MAPK signaling pathway (SP) contributes to aging and neurodegenerative diseases; however, information about its alteration in the retina with age and during AMD development is limited. To assess the contribution of alterations in p38 MAPK signaling to AMD, we compared age-associated changes in p38 MAPK SP activity in the retina between Wistar rats (control) and OXYS rats, which develop AMD-like retinopathy spontaneously. We analyzed changes in the mRNA levels of genes of this SP in the retina (data of RNA-seq) and evaluated the phosphorylation/activation of key kinases using Western blotting at different stages of AMD-like pathology including the preclinical stage. p38 MAPK SP activity increased in the retinas of healthy Wistar rats with age. The manifestation and dramatic progression of AMD-like pathology in OXYS rats was accompanied by hyperphosphorylation of p38 MAPK and MK2 as key p38 MAPK SP kinases. Retinopathy progression co-occurred with the enhancement of p38 MAPK-dependent phosphorylation of CryaB at Ser59 in the retina.

Enhanced cytotoxic activity of natural killer cells from increased calcium influx induced by electrical stimulation.

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Natural killer (NK) cells play a crucial role in immunosurveillance independent of antigen presentation, which is regulated by signal balance via activating and inhibitory receptors. The anti-tumor activity of NK cells is largely dependent on signaling from target recognition to cytolytic degranulation; however, the underlying mechanism remains unclear, and NK cell cytotoxicity is readily impaired by tumor cells. Understanding the activation mechanism is necessary to overcome the immune evasion mechanism, which remains an obstacle in immunotherapy. Because calcium ions are important activators of NK cells, we hypothesized that electrical stimulation could induce changes in intracellular Ca²⁺ levels, thereby improving the functional potential of NK cells. In this study, we designed an electrical stimulation system

and observed a correlation between elevated Ca^{2+} flux induced by electrical stimulation and NK cell activation. Breast cancer MCF-7 cells co-cultured with electrically stimulated KHYG-1 cells showed a 1.27-fold (0.5 V/cm) and 1.55-fold (1.0 V/cm) higher cytotoxicity, respectively. Electrically stimulated KHYG-1 cells exhibited a minor increase in Ca^{2+} level (1.31-fold (0.5 V/cm) and 1.11-fold (1.0 V/cm) higher), which also led to increased gene expression of granzyme B (GZMB) by 1.36-fold (0.5 V/cm) and 1.58-fold (1.0 V/cm) by activating Ca^{2+} -dependent nuclear factor of activated T cell 1 (NFAT1). In addition, chelating Ca^{2+} influx with 5 μM BAPTA-AM suppressed the gene expression of Ca^{2+} signaling and lytic granule (granzyme B) proteins by neutralizing the effects of electrical stimulation. This study suggests a promising immunotherapeutic approach without genetic modifications and elucidates the correlation between cytolytic effector function and intracellular Ca^{2+} levels in electrically stimulated NK cells.
